

Miriam Elia, *PhD*

Designing the systems behind trustworthy AI.



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Profile

Multidisciplinary scientist and systems thinker specializing in trustworthy AI. I design, evaluate, and help operationalize high-risk AI systems across their lifecycle, bridging technical, regulatory, clinical, and organizational contexts. My work translates emerging technologies into structured and accountable systems, through lifecycle design and human–AI collaboration. Based in Munich, I work internationally across Europe, with growing engagement across Southeast Asia. Open to relocation.

Career Path

- 2021 – 2026** **Doctor of Natural Sciences (PhD), University of Augsburg.** Chair of Software Methodologies for Distributed Systems. Defended *summa cum laude*, October 2025.
- › Designed *MQG4AI*, a Quality Gates methodology for Responsible AI lifecycle design, illustrated through certifiable AI use cases in medicine.
 - › Conceptualized and led *EsophagusVisualization*, an open-source clinical tool for interactive 3D-reconstructions from multi-modal esophageal imaging; coordinated an interdisciplinary team of computer science students, clinicians, and ethicists.
 - › Drove the *ECG-AI in Emergency Medicine* work with corpuls and the German Heart Center of TU Munich, with a focus on multi-label classification evaluation.
 - › Supervised five BSc and MSc theses and translated research to clinical, legal, and public audiences across Europe.
- 2019 – 2021** **M.Sc. Business Information Systems Engineering, University of Augsburg.** Research assistant at the Institute for Software & Systems Engineering: open-source *Decision Tree Learning* web application with international academic adoption (PMLR conference publication, 2020), and contributions to the Kognia research project.
- 2012 – 2019** **Studies and early industry experience.** B.Sc. Business Information Systems Engineering (2019) and B.Ed. English & French (2016), University of Augsburg. Backend development at Sigfox Germany GmbH (Munich, 2019); technical content and business development at Arrow Electronics (Frankfurt, 2017 – 2019).

Impact

- › *EsophagusVisualization* deployed at the University Hospital of Augsburg, adopted in active medical research
- › 6 first-author publications + 2 under review (full list at Google Scholar)
- › Best Position Paper Award, ICSoft 2023
- › 5 supervised BSc and MSc theses across AI & healthcare
- › AI literacy across patient, legal, and public audiences

Languages

German native **English** full professional **French** professional